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OPTICAL INSPECTION SYSTEM AUTOMATICALLY DEPLOYED ROVER

Abstract

An engine inspection apparatus includes a remotely controlled inspection rover, a rover deployment cart, and a stabilization tether having a first end attached to the inspection rover and a second end attached to the rover deployment cart. The stabilization tether is configured to prevent at least one of ingestion of the inspection rover by an engine while the engine is operating and expulsion of the inspection rover by the engine while the engine is operating.

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Background/Summary

TECHNICAL FIELD

[0001] This disclosure relates generally to engine inspection methods and apparatuses. More specifically, this disclosure relates to an optical inspection system automatically deployed rover.

BACKGROUND

[0002] The fan blades in a fan rotor stage of a gas turbine engine can be subject to damage caused by a variety of different factors. For example, foreign and/or domestic objects can strike a fan blade causing Foreign Object Damage (FOD) and Domestic Object Damage (DOD), surface erosion can occur, or vibratory response can produce high cycle fatigue that can lead to damage. Such damage may be randomly located (FOD/DOD) or it may be statistically more likely located in certain regions. For example, along the leading edge of a fan blade, and more specifically at certain regions of a fan blade leading edge. Current practices for inspecting fan blades involve a manual inspection of each fan blade by a technician when the turbine engine is shut down and cooled down. Inspections of this type are time intensive, costly, and require skilled technicians. There is also an elevated risk of FOD/DOD from the inspection itself from inspectors leaving behind objects that get ingested into the engine during the next engine start. More efficient and less costly processes that can be used to inspect fan blades are desirable.

SUMMARY

[0003] This disclosure provides methods and apparatus for an optical inspection system automatically deployed rover.

[0004] In one embodiment, an engine inspection apparatus is provided. The engine inspection apparatus includes a remotely controlled inspection rover, a rover deployment cart, and a stabilization tether having a first end attached to the inspection rover and a second end attached to the rover deployment cart. The stabilization tether is configured to prevent at least one of ingestion of the inspection rover by an engine while the engine is operating and expulsion of the inspection rover by the engine while the engine is operating.

[0005] The stabilization tether may include at least one of a power link configured to provide power to the inspection rover, a data reception link configured to receive control commands for operating the inspection rover, a data transmission link configured to transmit inspection data from the inspection rover, a coolant send line configured to provide a cooling medium to the inspection rover, a coolant return line configured to receive a cooling medium from the inspection rover, and a purge supply line configured to introduce a purging medium into an engine.

[0006] The inspection rover may include a frame, a locomotion system disposed on the frame, a stabilization locking mechanism configured to lock the frame on to a component of an engine, and at least one inspection sensor extending from the frame. The at least one inspection sensor may be a high-speed camera, an acoustical sensor, and/or a LIDAR. An inspection sensor positioning arm may be disposed between the at least one inspection sensor and the frame to position the at least one inspection sensor for performing an inspection operation.

[0007] The rover deployment cart may include a base, a locomotion system disposed on the base, and a rover deployment platform extending from the base. The rover deployment platform is configured to deploy the inspection rover into an engine housing while the engine is operating.

[0008] In another embodiment, a remotely controlled inspection rover is provided. The inspection rover includes a frame, a locomotion system disposed on the frame, a high-speed camera extending from the frame, and a stabilization tether having a first end attached to the inspection rover and a second end configured to be attached to a support apparatus. The support apparatus may be a rover deployment cart configured to deploy the inspection rover into an engine housing while the engine is operating. The inspection rover may further include a stabilization locking mechanism configured to lock the frame on to a component of an engine, and an inspection sensor positioning arm arranged between the high-speed camera and the frame configured to position the high-speed camera while performing an inspection operation.

[0009] In yet another embodiment, a method of operating an engine inspection apparatus is provided. The method includes deploying a remotely controlled inspection rover from a rover deployment cart into an engine housing while the engine is operating, maintaining force on a stabilization tether disposed between the inspection rover and the rover deployment cart while the

inspection rover is deployed in the engine housing, and performing an inspection operation with the inspection rover on a component of the engine, via an inspection sensor, while the engine is operating. Inspecting the component of the engine may include at least one of performing high speed photography of the component while the component is in motion, performing an acoustical analysis of the component while the component is in motion, and performing a LIDAR analysis of the component while the component is in motion. The component may include a plurality of fan blades. If the inspection rover is deployed into an inlet of the engine housing, maintaining force on the stabilization tether prevents the engine from ingesting the inspection rover. If the inspection rover is deployed into an exhaust of the engine housing, maintaining force on the stabilization tether prevents the engine from expelling the inspection rover. The temperature of the inspection rover may be regulated by a cooling medium. The cooling medium may be transported by the stabilization tether. The inspection rover may be locked to a component comprised by the engine prior to performing the inspection operation. The inspection sensor may be positioned by an inspection sensor positioning arm prior to performing the inspection operation. The inspection rover may be remotely controlled by receiving operation commands via the stabilization tether. The inspection rover may transmit inspection data related to the inspection operation. The inspection data may be transmitted via the stabilization tether.

[0010] Other technical features may be readily apparent to one skilled in the art from the following figures, descriptions, and claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] For a more complete understanding of this disclosure and its advantages, reference is now made to the following description, taken in conjunction with the accompanying drawings, in which:

[0012] FIG. 1 illustrates a block diagram for an example engine inspection apparatus according to embodiments of the present disclosure;

[0013] FIG. 2 illustrates an example inspection rover according to embodiments of the present disclosure;

[0014] FIG. 3 illustrates an example of an engine inspection apparatus in use according to embodiments of the present disclosure;

[0015] FIG. 4 is a view of an example turbine engine according to embodiments of the present disclosure; and

[0016] FIG. 5 illustrates a method of operating an engine inspection apparatus according to embodiments of the present disclosure.

DETAILED DESCRIPTION

[0017] FIGS. 1 through 5, discussed below, and the various embodiments used to describe the principles of this disclosure in this patent document are by way of illustration only and should not be construed in any way to limit the scope of the disclosure. Those skilled in the art will understand that the principles of this disclosure may be implemented for any suitably arranged engine inspection apparatus.

[0018] One potential solution that can be used to inspect fan blades is an Optical Inspection System (OIS), which may be deployed while the engine is operating, and a processing system to determine if any appreciable Foreign Object Damage (FOD) is present. The determination may lead to a pass/fail decision and potentially initiate a manual inspection of the fan blades. In one embodiment, the OIS sends image data from a camera, for example, to the Full Authority Digital Engine Control (FADEC) for processing/storage and the image data can be offloaded for review for disposition on required maintenance and inspection.

[0019] In one embodiment of an OIS, an engine-embedded camera is deployed for image capture

and stowed during normal operation. However, this embodiment has many challenges. One is the need for very low rotational speeds during image capture due to the lower Frames Per Second (FPS) capability of the small camera needed for engine case integration. Another is the overall integration with the engine itself, e.g., controls, cooling/purge, data processing/storage, etc. Also, if the OIS is integrated with the engine, it must survive all operating conditions even if the OIS is only in-use when aircraft is on the ground. With future programs there might be a need to replace manual inspections, referred to as “duct dives” altogether due to smaller inlets. Even with the signal from engine-embedded OIS requiring a duct dive, if a person doesn't fit, only a borescope inspection ability is available.

[0020] One alternative to a case-integrated OIS is a fully detached remotely controlled (RC) rover that can be deployed into an inlet of an engine housing of, for example an aircraft, while the aircraft is on the ground. In one embodiment, the rover is configured to fit within a variety of inlet and exhaust duct sizes and shapes. In one embodiment, the rover is configured to carry at least one inspection sensor. The inspection sensor may be, for example, a very capable high-speed camera. While the cost of a rover may be more than an integrated camera, overall cost for a fleet of engines may be lower, as fewer cameras may be required. For example, a single rover may be used to inspect multiple engines. Furthermore, the rover may be configured to inspect multiple types of engines, bringing further economy to an engine fleet.

[0021] Because the rover is completely detached from the engine, all data processing/storage, purge/cooling, etc., can be handled without additional per unit cost to the aircraft/engine. In one embodiment the rover is manually deployed by a maintainer. In another embodiment, a fully-automatic deployment of the rover is commanded by the FADEC. Such embodiments provide numerous other advantages. For example, no engine-mounted sensors/interfaces are needed for the OIS capability. Multiple fields of view may be possible, as the rover may be configured to carry multiple cameras and/or lights, be configured to reposition the camera(s), etc. In one embodiment, the rover is configured without a FADEC interface or any other engine control.

[0022] Additionally, as the rover can perform the inspection without a human duct dive, the inspection may be performed while the engine is operating, while the engine is at high temperature, etc. This enables faster inspection turnaround, as the engine may be inspected, for example, as soon as an aircraft lands, without shutting down the engine.

[0023] While the rover is described herein as being remotely controlled, in some embodiments the remote control may be manual remote control. For example, a technician may control the rover via a human interface such as a joystick, a touchscreen, etc. In other embodiments, the remote control may be fully autonomous. For example, the rover may be controlled by an artificial intelligence system, a previously programmed automation system, etc. In some embodiments, the control may be a hybrid between manual control and autonomous control.

[0024] FIG. 1 illustrates a block diagram for an example engine inspection apparatus **100** according to embodiments of the present disclosure. The embodiment of engine inspection apparatus **100** illustrated in FIG. 1 is for illustration only. Different embodiments of engine inspection apparatus **100** could be used without departing from the scope of this disclosure.

[0025] In the example of FIG. 1, engine inspection apparatus **100** includes an inspection rover **110** and a rover deployment cart **130**. Rover deployment cart **130** is joined to inspection rover **110** by a stabilization tether **150**. In circumstances where inspection rover **110** is deployed into an engine housing while the engine is operating, applying a force to stabilization tether **150** helps avoid FOD to the engine by inspection rover **110** while inspection rover **110** is deployed. For example, when inspection rover **110** is deployed into an inlet, tension may be maintained on stabilization tether **150** between inspection rover **110** and rover deployment cart **130** to prevent inspection rover **110** from being ingested into the engine. In another example, when inspection rover **110** is deployed into an exhaust, a stiffening force may be maintained on stabilization tether **150** between inspection rover **110** and rover deployment cart **130** to prevent inspection rover **110** from being expelled from

the exhaust.

[0026] While stabilization tether **150** is primarily used to prevent FOD and stabilize the position of inspection rover **110** within an engine housing while the engine is operating, in some embodiments stabilization tether may provide additional support capabilities for inspection rover **110** and/or rover deployment cart **130**. For example, FIG. **1** illustrates that stabilization tether **150** comprises support links **150a-150f**. However, it should be understood that in some embodiments, any number of support links **150a-150f** may be omitted, and that any number of additional support links may be included in other embodiments.

[0027] Support links **150a-150f** may be representative of any of a number of support functions. Examples of support functions that may be provided by one of support links **150a-150f** include, but are not limited to: [0028] providing electrical power to inspection rover **110** [0029] providing control communication to inspection rover **110** [0030] providing data communications to and from inspection rover **110** [0031] circulating a temperature regulating medium for inspection rover **110** [0032] supplying purge air to an engine under inspection

[0033] For example, support link **150a** may be power supply path (e.g., positive voltage) for providing power to inspection rover **110**, and support link **150b** may be a power return path (e.g., ground) for completing the power supply circuit for inspection rover **110**. In another example, support link **150c** may be a coolant send line for providing a cooling medium to inspection rover **110** to assist in maintaining inspection rover **110**'s temperature. Similarly, support link **150d** may be a coolant return line for transporting the cooling medium away from inspection rover **110**. For instance, the cooling medium may be a liquid coolant, and support links **150c** and **150d** may form a closed coolant loop between inspection rover **110** and an external radiator and coolant pump system. In another example, the coolant medium may be chilled air, and support links **150c** and **150d** may form an open loop with inspection rover **110** where chilled air is blown into support link **150c** and exhausted through **150d** to a point external from the engine housing. In other example, support link **150e** may provide control commands to inspection rover **110** so that inspection rover **110** maneuvers within the engine housing, performs various inspection operations, etc. In yet another example, support link **150f** may provide a data path for inspection rover **110** to transmit inspection data. For example, inspection rover **110** may use support link **150f** to transmit high-speed camera images that may be stored, processed, etc. by rover deployment cart **130** or some other external system. Because stabilization tether **150** may provide these support functions to inspection rover **110**, inspection rover **110** has few limitations with regards to electrical power, purge-supply, cooling capacity, data rate, etc.

[0034] While FIG. **1** illustrates engine inspection apparatus **100** as including a single inspection rover **110** and a single stabilization tether **150**, it should be understood that in some embodiments engine inspection apparatus **100** may include multiple inspection rovers **110** and/or multiple stabilization tethers **150**. In some embodiments, inspection apparatus **100** may include multiple rover deployment carts **130**.

[0035] While FIG. **1** illustrates engine inspection apparatus **100** as including inspection rover **110**, rover deployment cart **130**, and stabilization tether **150**, it should be understood that some embodiments may omit at least one of rover deployment cart **130** and/or stabilization tether **150**. For example, in some embodiments, inspection rover **110** may be an entirely self-contained system. In a self-contained system, rover **110** may include on-board data processing and storage, as well as wireless data communication capabilities. In other embodiments, inspection rover **110** may be deployed by rover deployment cart **130** without a stabilization tether (e.g., while the engine is shut down). In such embodiments, inspection rover **110** may be designed with a particular aerodynamic shape (e.g., an airfoil) to help avoid ingestion if deployed while the engine is in operation. In some embodiments, inspection rover **110** may be deployed manually, and stabilization tether **150** may be attached to an apparatus other than rover deployment cart **130**. For example, a technician may place inspection rover **110** within the engine inlet, and stabilization tether **150** may be attached to a fixed

point (e.g., a pole), a non-mobile support apparatus etc.

[0036] Although FIG. 1 illustrates one example of a block diagram for an example engine inspection apparatus **100**, various changes may be made to FIG. 1. For example, the number of support links may change, the rover deployment cart could support multiple rovers, etc.

[0037] FIG. 2 illustrates an example inspection rover **110** according to embodiments of the present disclosure. The embodiment of inspection rover **110** illustrated in FIG. 2 is for illustration only. Different embodiments of inspection rover **110** could be used without departing from the scope of this disclosure.

[0038] In the example of FIG. 2, inspection rover **110** includes a frame **112**. Frame **112** provides mechanical support for inspection rover **110**. For example, in some embodiments, stabilization tether **150** may attach to frame **112** to prevent ingestion or expulsion of inspection rover **110** from an engine housing. A locomotion system **114** is disposed on frame **112**. Locomotion system **114** enables inspection rover **110** to position itself within a particular area of an engine housing. For example, locomotion system **114** may enable inspection rover **110** to drive within an intake to a particular location within the engine housing, so that a particular engine component may be inspected (e.g., a fan blade). Because inspection rover **110** may be controlled remotely, locomotion system **114** may be utilized to position inspection rover **110** without the need for a human duct dive within the engine housing. While locomotion system **114** is illustrated as a plurality of motorized tank treads, it should be understood that any locomotion system may be utilized. For example, locomotion system **114** may be implemented as motor driven wheels, an aerodynamic drive (e.g., a fan, propeller, etc.), walking legs, etc., according to particular needs. In some embodiments, locomotion system **114** may be omitted. For example, inlet flow from an operating engine under inspection may be used to draw inspection rover **110** within the engine housing of the engine under inspection.

[0039] In the example of FIG. 2, inspection rover **110** further includes an inspection sensor **116** extending from frame **112**. Inspection sensor **116** is utilized to inspect engine components. For example, inspection sensor **116** may be utilized to inspect fan blades on a turbine engine within an engine housing of an aircraft while the engine is operating. For example, inspection sensor **116** may be a high-speed camera. However, inspection sensor **116** may be any type of sensor suitable for inspecting a particular component. For example, inspection sensor **116** could be an acoustical sensor, such as an ultrasonic sensor or a microphone for ping testing, a LIDAR imaging system, a borescope arm enabling Rotor 1 Trailing Edge (R1TE) and beyond images, a magnetic sensor, etc. In some embodiments, inspection sensor **116** is modular and may be interchanged with other inspection sensors. While FIG. 2 only depicts a single inspection sensor **116**, it should be understood that inspection rover **110** may be configured to employ multiple inspection sensors **116** simultaneously. For example, inspection rover **110** may be configured with multiple sensors of different types, so that multiple modes of inspection may be performed simultaneously or within a single deployment. In another example, inspection rover **110** may be configured with multiple sensors (e.g., cameras) of the same type, so that inspection may be performed at different angles, etc., simultaneously (e.g., capturing images from different camera views without repositioning the camera or the rover).

[0040] In the example of FIG. 2, inspection rover **110** further includes an accessory **118**. Accessory **118** may provide any functionality that can assist the rover during deployment. For example, accessory **118** may be a stabilization locking mechanism. In this example, a stabilization locking mechanism refers to an apparatus that locks inspection rover **110** to e.g., an engine component, to stabilize and/or secure inspection rover **110** while performing an inspection operation. For example, accessory **118** could lock inspection rover **110** to either one or both of fan inlet variable vanes (FIVVs) **420** and **422** illustrated in FIG. 4. While FIG. 2 depicts a single accessory **118**, it should be understood that inspection rover **110** may be configured with any number of accessories **118**, or accessory **118** may be omitted. In some embodiments, accessory **118** is modular and may be

interchanged with other accessories. For example, in addition to being a stabilization locking mechanism, accessory **118** may be any of an additional inspection sensor, a repair tool such as a blending apparatus, a nozzle which may apply coatings such as radar mitigation, anti-icing, etc. In embodiments that employ a nozzle, the nozzle may be fed via stabilization tether **150** via a support link such as one of support links **150a-150f**.

[0041] In the example of FIG. 2, inspection rover **110** further includes an inspection sensor positioning arm **120**. Positioning arm may be used to position inspection sensor **116** without repositioning inspection rover **110**. For example, a particular component may be located at a position that cannot be easily viewed when inspection rover **110** is located at a particular location within an engine housing. Positioning arm **120** for example may provide pan, tilt, elevation, etc. capability for inspection sensor **116**. However, it should be understood that in some embodiments, positioning arm **120** may be omitted. In such embodiments, inspection sensor **116** may be disposed directly on frame **112**, and position sensor **116** may be repositioned by repositioning of inspection rover **110**.

[0042] Although FIG. 2 illustrates one example of an example inspection rover **110**, various changes may be made to FIG. 2. For example, the number of inspection sensors may change, the number of accessories may change, a different type of locomotion system could be used, etc.

[0043] FIG. 3 illustrates an example **300** of engine inspection apparatus **100** in use according to embodiments of the present disclosure. The embodiment of engine inspection apparatus **100** illustrated in FIG. 3 is for illustration only. Different embodiments of engine inspection apparatus **100** could be used without departing from the scope of this disclosure.

[0044] In the example of FIG. 3, engine inspection apparatus **100** is shown configured with inspection rover **110** and rover deployment cart **130**. While FIG. 3 does not depict stabilization tether **150**, it should be understood that in the example of FIG. 3, inspection apparatus **100** may be configured with stabilization tether **150**. In the example of FIG. 3, rover deployment cart **130** is deploying inspection rover **110** into an inlet **304** of an aircraft **302**. For example, aircraft **302** may comprise an engine **402** which is illustrated in FIG. 4.

[0045] To assist in deploying inspection rover **110**, rover deployment cart **130** includes a base **132**, a locomotion system **134** disposed on base **132**, and a rover deployment platform **136** extending from base **132**. Rover deployment platform **136** is configured to deploy inspection rover **110** into an engine housing while the engine is operating. For example, rover deployment platform **136** may raise or lower inspection rover **110** such that inspection rover **110** may drive into inlet **304** of aircraft **302**. Rover deployment cart **130** may further support rover **110** by supplying power, data storage, data analysis, etc. via stabilization tether **150**, providing an anchoring point for stabilization tether **150**, etc. Locomotion system **134** enables deployment cart **130** to position itself to deploy inspection rover **110**. For example, deployment cart **130** could be positioned via remote control manually by a technician, be positioned automatically via an artificial intelligence system, be positioned by a preprogrammed automation system, etc. While locomotion system **134** is illustrated as a plurality of motorized wheels, it should be understood that any locomotion system may be utilized. For example, locomotion system **114** may be implemented as motor driven treads, walking legs, an air lift, etc., according to particular needs.

[0046] Although FIG. 3 illustrates one example **300** of engine inspection apparatus **100** in use, various changes may be made to FIG. 3. For example, the rover deployment cart could support multiple rovers, an inlet for a different type of apparatus than an aircraft (e.g., a generator) could be shown, etc.

[0047] FIG. 4 illustrates views **400a-400d** of an example turbine engine **402** according to embodiments of the present disclosure. The embodiment of engine **402** illustrated in FIG. 4 is for illustration only. Different embodiments of an engine **402** could be used without departing from the scope of this disclosure.

[0048] In the example of FIG. 4, views **400a-400d** depict turbine engine **402**. View **400a** is a

forward looking aft view of turbine engine **402**. View **400b** is an isometric view of turbine engine **402**. Views **400c** and **400d** are illustrative of a standard field of view of inspection sensor **116** while engine inspection apparatus **100** is performing an inspection of turbine engine **402**. In some embodiments, inspection apparatus **100** may be calibrated based on a particular field of view, such as one of views **400c** and/or **400d**, etc. Turbine engine **402** includes fan inlet variable vane (FIVV) struts **420** and **422**. Turbine engine **402** may be installed in an engine housing of a different apparatus, such as aircraft **302**, a generator, a ship, etc. During an inspection, a remotely controlled inspection rover, such as inspection rover **110** may use an accessory such as a stabilization locking mechanism to lock itself to engine **402**. For example, inspection rover **110** may utilize accessory **118** to lock itself to one or both of FIVV struts **420** and **422** and/or another component. In this manner, inspection rover **110** may perform inspection operations while engine **402** is operating. For example, locking on to engine **402** may assist in preventing FOD to engine **402** from inspection rover **110**, help prevent ingestion of inspection rover **110**, may stabilize the position of inspection rover **110** while performing inspection operations, etc.

[0049] Although FIG. **4** illustrates one example of a turbine engine **402** in use, various changes may be made to **4**. For example, engine **402** could be a different type of engine, could include different components, etc.

[0050] FIG. **5** illustrates a method **500** of operating an engine inspection apparatus according to embodiments of the present disclosure. An embodiment of the method illustrated in FIG. **5** is for illustration only. One or more of the components illustrated in FIG. **5** may be implemented in specialized circuitry configured to perform the noted functions or one or more of the components may be implemented by one or more processors executing instructions to perform the noted functions. Other embodiments of a method **500** of operating an engine inspection apparatus could be used without departing from the scope of this disclosure.

[0051] As illustrated in FIG. **5**, the method **500** begins at step **502**. At step **502**, a remotely controlled inspection rover, such as inspection rover **110** of FIG. **1**, is deployed into an engine housing while the engine is operating. For example, the inspection rover may be deployed from a rover deployment cart such as rover deployment cart **130** of FIG. **1**. At step **504**, a force is maintained on a stabilization tether, such as stabilization tether **150** of FIG. **1**, while the inspection rover is deployed in the engine housing. The stabilization tether may be disposed between the inspection rover and the rover deployment cart. At step **506**, the temperature of the inspection rover is regulated. For example, the stabilization tether may transport a cooling medium to and from the inspection rover. At step **508**, the inspection rover is locked to an engine component. For example, a stabilization locking mechanism disposed on the inspection rover may be locked to a component of the engine. At step **510**, an inspection sensor, such as inspection sensor **116** of FIG. **2** is positioned. For example, the inspection sensor may be positioned by a positioning arm, such as positioning arm **120** of FIG. **2**, or the inspection rover may position itself with a locomotion system to position the inspection sensor. At step **512**, an inspection operation is performed with the inspection rover while the engine is running. For example, the inspection rover may inspect a component of the engine, such as a plurality of fan blades, while the component is in motion. The inspection may be one of high-speed photography, an acoustical analysis (e.g., a ping test), a LIDAR analysis, etc. Finally, at step **514**, the inspection rover transmits inspection data related to the inspection operation. For example, the inspection data may be transmitted via the stabilization tether, a wireless transmission system such as WiFi or Bluetooth, etc.

[0052] Although FIG. **5** illustrates one example of a method **500** of operating an engine inspection apparatus, various changes may be made to FIG. **5**. For example, while shown as a series of steps, various steps in FIG. **5** could overlap, occur in parallel, occur in a different order, or occur any number of times.

[0053] In some embodiments, various functions described in this patent document are implemented or supported by a computer program that is formed from computer readable program code and that

is embodied in a computer readable medium. The phrase “computer readable program code” includes any type of computer code, including source code, object code, and executable code. The phrase “computer readable medium” includes any type of medium capable of being accessed by a computer, such as read only memory (ROM), random access memory (RAM), a hard disk drive, a compact disc (CD), a digital video disc (DVD), or any other type of memory. A “non-transitory” computer readable medium excludes wired, wireless, optical, or other communication links that transport transitory electrical or other signals. A non-transitory computer readable medium includes media where data can be permanently stored and media where data can be stored and later overwritten, such as a rewritable optical disc or an erasable storage device.

[0054] It may be advantageous to set forth definitions of certain words and phrases used throughout this patent document. The terms “application” and “program” refer to one or more computer programs, software components, sets of instructions, procedures, functions, objects, classes, instances, related data, or a portion thereof adapted for implementation in a suitable computer code (including source code, object code, or executable code). The term “communicate,” as well as derivatives thereof, encompasses both direct and indirect communication. The terms “include” and “comprise,” as well as derivatives thereof, mean inclusion without limitation. The term “or” is inclusive, meaning and/or. The phrase “associated with,” as well as derivatives thereof, may mean to include, be included within, interconnect with, contain, be contained within, connect to or with, couple to or with, be communicable with, cooperate with, interleave, juxtapose, be proximate to, be bound to or with, have, have a property of, have a relationship to or with, or the like. The phrase “at least one of,” when used with a list of items, means that different combinations of one or more of the listed items may be used, and only one item in the list may be needed. For example, “at least one of: A, B, and C” includes any of the following combinations: A, B, C, A and B, A and C, B and C, and A and B and C.

[0055] The description in the present disclosure should not be read as implying that any particular element, step, or function is an essential or critical element that must be included in the claim scope. The scope of patented subject matter is defined only by the allowed claims. Moreover, none of the claims invokes 35 U.S.C. § 112(f) with respect to any of the appended claims or claim elements unless the exact words “means for” or “step for” are explicitly used in the particular claim, followed by a participle phrase identifying a function. Use of terms such as (but not limited to) “mechanism,” “module,” “device,” “unit,” “component,” “element,” “member,” “apparatus,” “machine,” “system,” “processor,” or “controller” within a claim is understood and intended to refer to structures known to those skilled in the relevant art, as further modified or enhanced by the features of the claims themselves, and is not intended to invoke 35 U.S.C. § 112(f).

[0056] While this disclosure has described certain embodiments and generally associated methods, alterations and permutations of these embodiments and methods will be apparent to those skilled in the art. Accordingly, the above description of example embodiments does not define or constrain this disclosure. Other changes, substitutions, and alterations are also possible without departing from the spirit and scope of this disclosure, as defined by the following claims.

Claims

1. An engine inspection apparatus comprising: a remotely controlled inspection rover; a rover deployment cart; and a stabilization tether having a first end attached to the inspection rover and a second end attached to the rover deployment cart.
2. The engine inspection apparatus of claim 1, wherein the stabilization tether is configured to prevent at least one of: ingestion of the inspection rover by an engine while the engine is operating; and expulsion of the inspection rover by the engine while the engine is operating.
3. The engine inspection apparatus of claim 1, wherein the stabilization tether comprises at least one of: a power link configured to provide power to the inspection rover; a data reception link

configured to receive control commands for operating the inspection rover; a data transmission link configured to transmit inspection data from the inspection rover; a coolant send line configured to provide a cooling medium to the inspection rover; a coolant return line configured to receive a cooling medium from the inspection rover; and a purge supply line configured to introduce a purging medium into an engine.

4. The engine inspection apparatus of claim 1, wherein the inspection rover comprises: a frame; a locomotion system disposed on the frame; a stabilization locking mechanism configured to lock the frame on to a component of an engine; and at least one inspection sensor extending from the frame.

5. The engine inspection apparatus of claim 4, wherein the at least one inspection sensor is at least one of: a high-speed camera; an acoustical sensor; and a LIDAR.

6. The engine inspection apparatus of claim 4, further comprising, an inspection sensor positioning arm, disposed between the at least one inspection sensor and the frame, configured to position the at least one inspection sensor for performing an inspection operation.

7. The engine inspection apparatus of claim 1, wherein the rover deployment cart comprises: a base; a locomotion system disposed on the base; and a rover deployment platform extending from the base, wherein the rover deployment platform is configured to deploy the inspection rover into an engine housing while the engine is operating.

8. A remotely controlled inspection rover comprising: a frame; a locomotion system disposed on the frame; a high-speed camera extending from the frame; and a stabilization tether having a first end attached to the inspection rover and a second end configured to be attached to a support apparatus.

9. The inspection rover of claim 8, wherein the support apparatus is a rover deployment cart configured to deploy the inspection rover into an engine housing while the engine is operating.

10. The inspection rover of claim 8, further comprising: a stabilization locking mechanism configured to lock the frame on to a component of an engine; and an inspection sensor positioning arm arranged between the high-speed camera and the frame configured to position the high-speed camera while performing an inspection operation.

11. A method of operating an engine inspection apparatus comprising: deploying a remotely controlled inspection rover from a rover deployment cart into an engine housing, while the engine is operating; maintaining force on a stabilization tether disposed between the inspection rover and the rover deployment cart while the inspection rover is deployed in the engine housing; and performing an inspection operation with the inspection rover on a component of the engine, via an inspection sensor, while the engine is operating.

12. The method of claim 11, wherein inspecting the component of the engine comprises at least one of: performing high speed photography of the component while the component is in motion; performing an acoustical analysis of the component while the component is in motion; and performing a LIDAR analysis of the component while the component is in motion.

13. The method of claim 11, wherein the component comprises a plurality of fan blades.

14. The method of claim 11, wherein the inspection rover is deployed into an inlet of the engine housing, and maintaining force on the stabilization tether prevents the engine from ingesting the inspection rover.

15. The method of claim 11, wherein the inspection rover is deployed into an exhaust of the engine housing, and maintaining force on the stabilization tether prevents the engine from expelling the inspection rover.

16. The method of claim 11, further comprising regulating a temperature of the inspection rover via a cooling medium transported by the stabilization tether.

17. The method of claim 11, further comprising locking the inspection rover to a component comprised by the engine prior to performing the inspection operation.

18. The method of claim 11, further comprising positioning the inspection sensor via an inspection sensor positioning arm prior to performing the inspection operation.

19. The method of claim 11, further wherein the inspection rover is remotely controlled by receiving operation commands via the stabilization tether.

20. The method of claim 11, further comprising, transmitting inspection data related to the inspection operation from the inspection rover, wherein the inspection data is transmitted via the stabilization tether.
